

WORKING DRAFT

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Paddy Value Chain Opportunities for Balaghat



Discussion Document

February 2009

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Executive Summary (1/2)

In Balaghat District (Orissa), Paddy is a critical crop for both food security and income of small and marginal farmers

- Paddy is grown by over 1.58 lakhs families (of which 1.35 lakhs are small and marginal) in Balaghat and contributes to the assets and/or income of some 14.98 lakh population (80% of rural population)
- Paddy is also the single largest crop acreage in Balaghat, representing 74% of total sown area (249,811 ha)
- Paddy represents ~30% of the calories in the Indian staple diet, playing a critical food security role in a district where the BPL population 146,094
- Balaghat abundant rain and has mostly medium and low paddy land. Its agro climatic conditions provide it with a source of competitive advantage relative to other Indian states – thereby creating the potential for paddy to act as an engine of growth through production and/or value addition

In spite of its potential, most small and marginal paddy farmers live in poverty due to poor productivity and price realization

- Balaghat paddy farmers currently achieve low yields (2.3 t/ha) due to: erratic rainfall, lack of knowledge / skills, accessible / affordable quality inputs, credit as well as land degradation
- In spite of its potential, Balaghat is a net 2.311 lakh ton paddy importer, and over the next 10 years there will be an additional 0.67 lakh MT required resulting in a 2.98 MT deficit; Meeting this demand would require a 172% improvement in yield or an additional 1.79 lakh hectares. Increasing paddy area is difficult, but yield improvement is definitely possible .

Executive Summary (2/2)

Two focused interventions should enable producers in Balaghat to achieve 400% of the productivity potential, resulting in a ~400% increase in profitability for farmers

- We have identified over 8 interventions to improve the paddy value chain – where the majority of the value comes from the introduction of SRI and/or targeted variety selection and direct linkages with FCI/millers
- In the event that direct linkages are not achievable, there are X additional interventions that can be pursued at a community level to improve their paddy productivity and profitability
 - [List interventions – What is it? Value? How could you do it? Issues/Questions?]
- In an optimized paddy production system in Balaghat, there are opportunities to improve yields by up to 140%, which would bring Balaghat up to 200% of the national average

To capture these benefits for the paddy farmers of Balaghat , we propose next steps

- Increase yields through better inputs
- Increase Irrigation
- SRI + better input + irrigation
- Direct sales to market
- Cleaning, drying, and grading
- Sell in off season
- Transparent weighing system

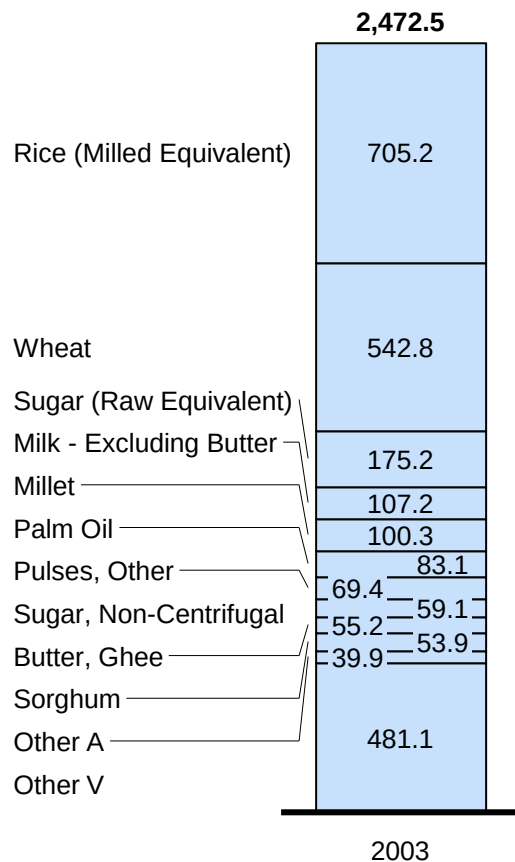
this text is not complete – need to complete it as per the document; Also, next step shouldn't be just to list interventions... is there a subset of priority interventions that should be done over the short term, and then other things for medium term? Any additional questions that need to be addressed?

Today's discussion

- **Why is Paddy a critical crop for both India and Balaghat?**
- What are the current challenges and potential end state?
- What are the potential interventions?
- What should we do next?

Rice is a critical crop for the food security of India, representing ~30% of per capita calories consumed in a country where ~30% of the population is hungry and malnourished

Total calorie consumption per capita



- Rice share of local diet as remained fairly consistent at 30-35% of total calories consumed per person (~81.1 kgs of rice per person) since the 1960s reflecting its role as a primary food grain for most Indians
- Although at least 30% of rice is consumed on the farm, an estimated 60m farmers use paddy as a source of savings / income

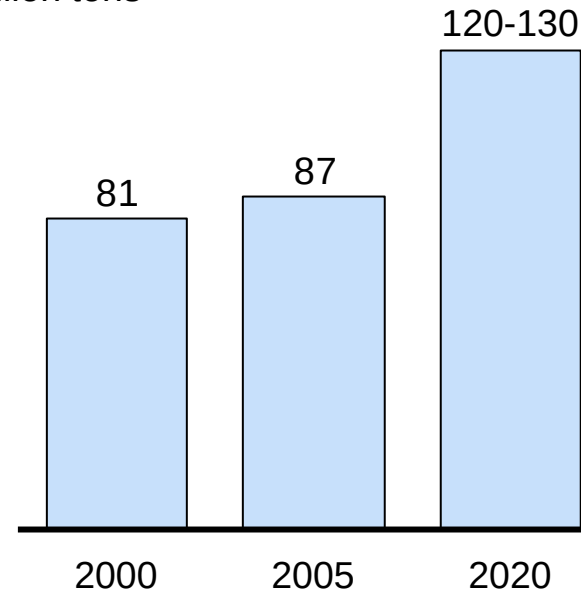
Rice demand is expected to grow at 2% per annum through 2020 based on rising income levels and increased urbanization

Role of Rice

1. Rice is an important cereal and part of the Indian staple diet across several regions and income levels
2. Rice consumption is unlikely to be displaced by any other food category
3. Rice, being a fine grain, will be consumed in a higher quantity than coarse cereals (such as Bajra and Jowar) with rising income levels
4. There is a currently a huge 'unmet' demand with an estimated 100 million households eating less than required

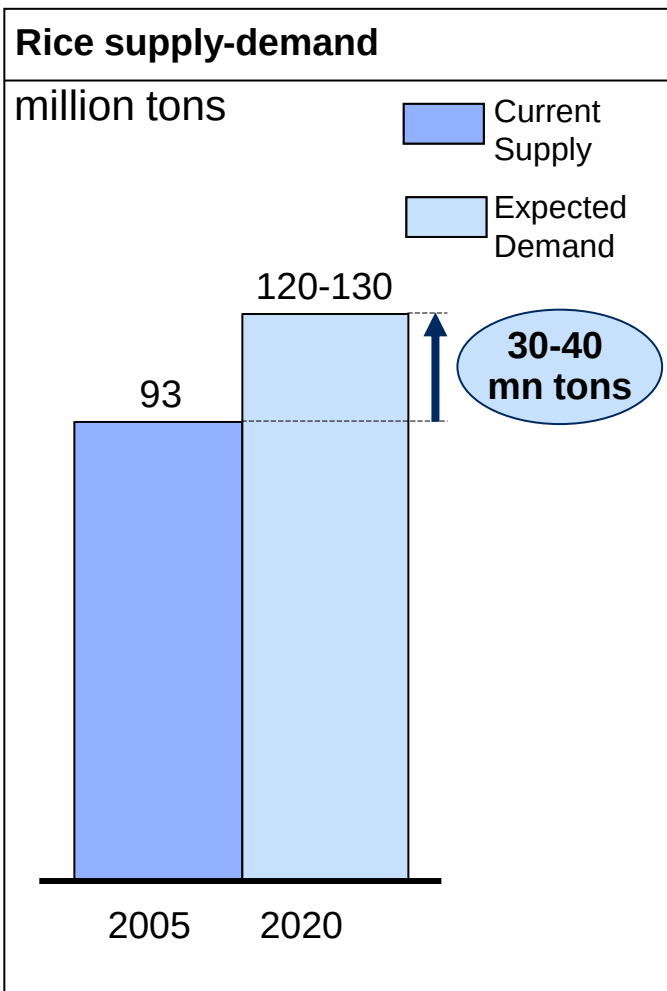
Rice demand

million tons



- Historical growth rates of rice demand were about 1.5% annually
- The demand growth is expected to accelerate to over 2% annually due to rising income levels and increased urbanization which leads to a change in consumption patterns

Meeting this increased demand will require a 30% improvement in yields given that import markets are thin and there is limited acreage that can be profitably shifted to Paddy cultivation



Local production critical

No depth in global rice market

- Total global market is ~15 MT (2007)
- India is a net exporter of rice when domestic production is in surplus

Global trade is highly volatile

- International trade has been thin as exporters have demonstrated high willingness to curb trade in the interest of meeting local demand
 - Thailand banned exports
 - Vietnam is reconsidering exports

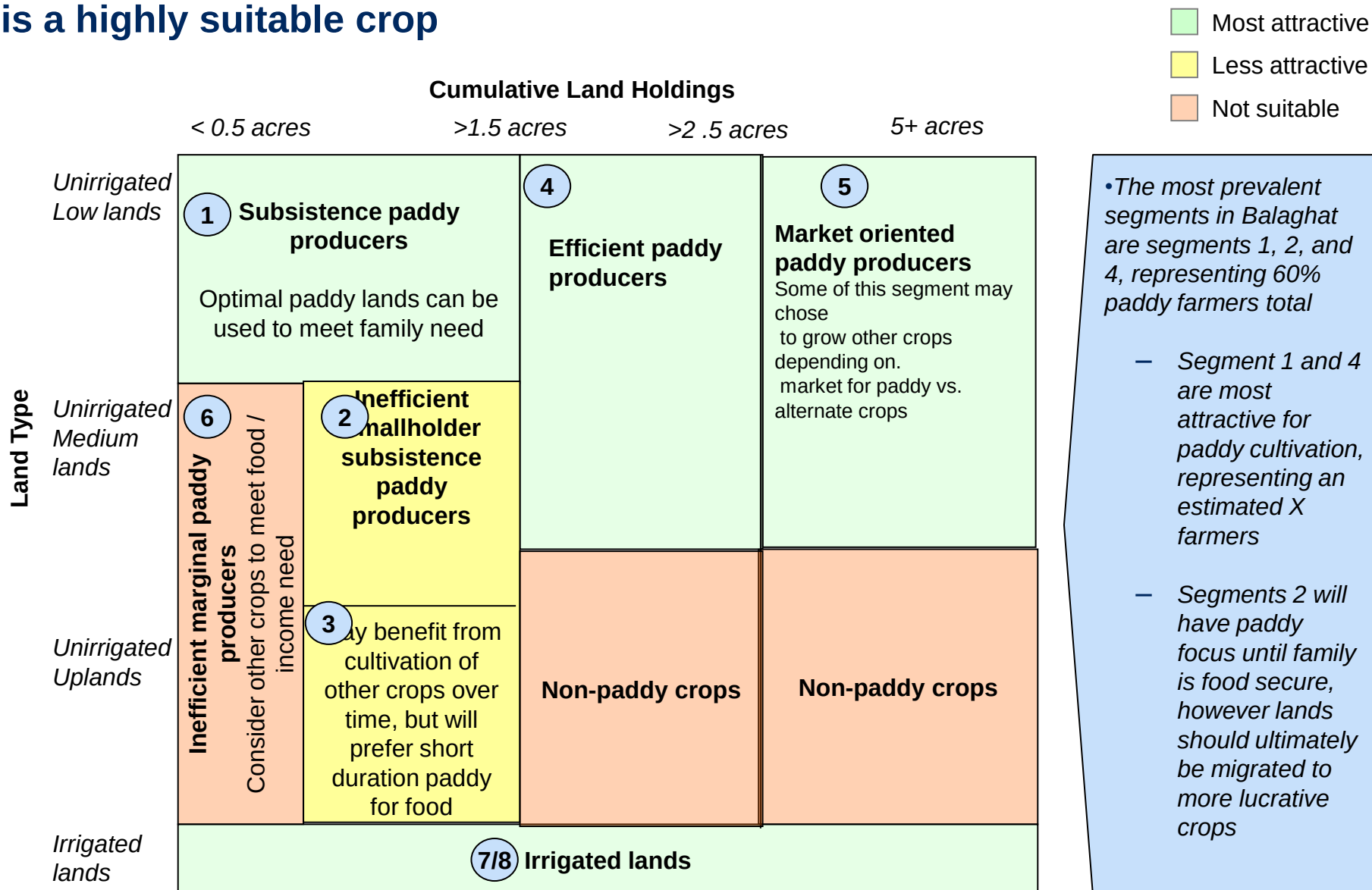
Transport and storage of rice is an issue

- Rice needs to be transported and stored under specific conditions which complicates global trade
- Current infrastructure (ports and warehouses) in India are not adequate for high quantum of rice imports

Today's discussion

- Why is Paddy a critical crop for both India and Balaghat?
- **What are the current challenges and potential end state?**
- What are the potential interventions?
- What should we do next?

If we consider the suitability of Paddy for Balaghat farmers, we find that it is a highly suitable crop



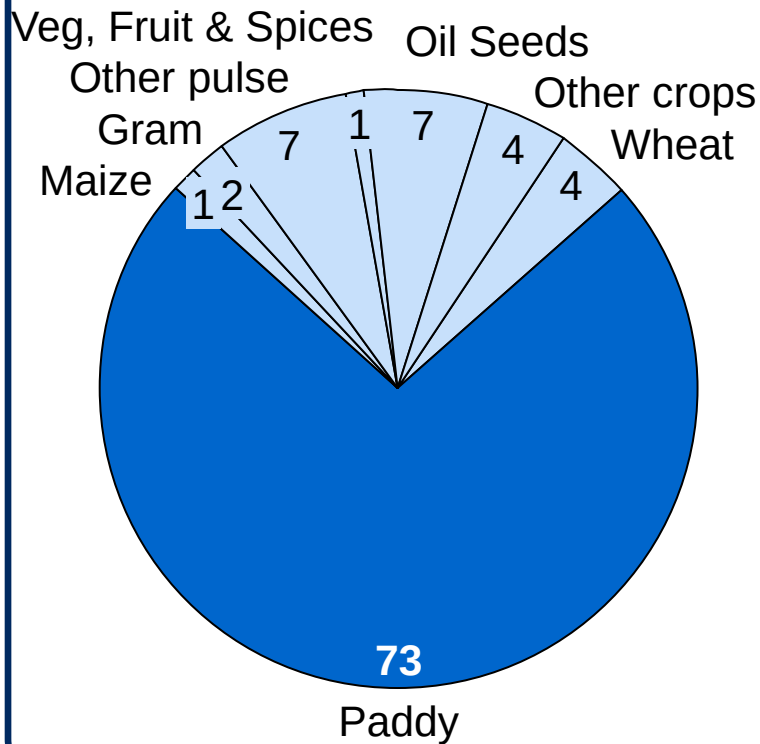
Balaghat has a highly suitable climate for paddy cultivation, significant bunded acreage and appropriate climate for optimal for paddy production

- **Balaghat has highly suitable land for paddy cultivation:**
 - Low and medium lands have been converted in paddy land by bunding
 - Bunding has also enabled farmers to harvest sufficient water for paddy cultivation in Kharif
 - Given that there is so much paddy land, it is most efficient to use all lowlands for paddy cultivation as the growing of other crops in between paddy patches are difficult due to high moisture
- **Balaghat has a highly suitable climate for paddy cultivation:**
 - Rice being a tropical and sub-tropical plant, requires a fairly high temperature, ranging from 20° to 40°C which is precisely the range in Balaghat

Paddy is already the largest acreage so farmers have interest and experience in cultivation

Total Acreage by Crop

Percent, 100% = X,XXX acres

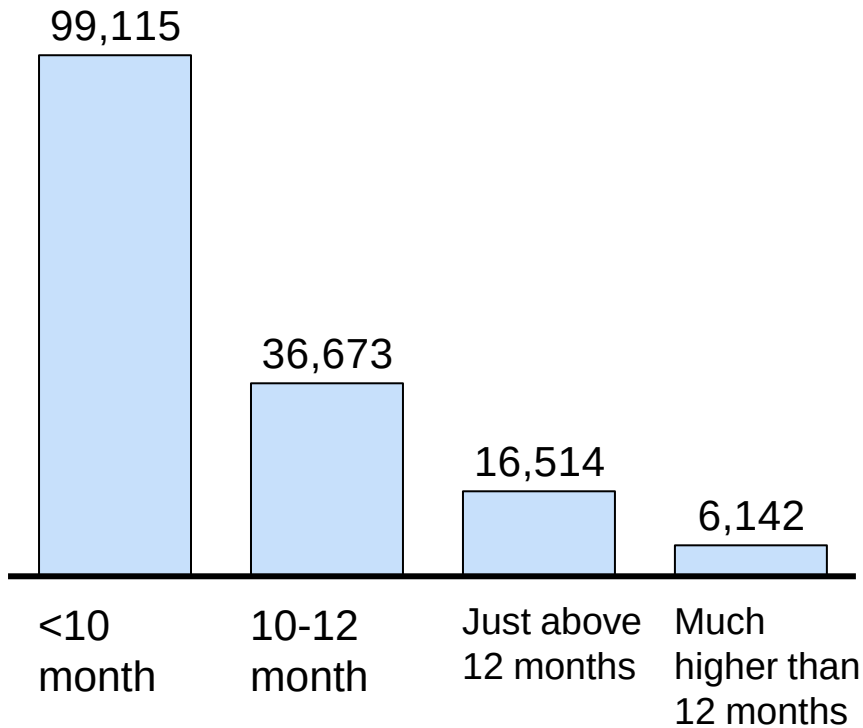


- Paddy is the dominant Kharif crop, representing 74% of total cropped area in Balaghat given its critical role in food security in the region**
 - Over 75% of households in Balaghat (2.13 lakh families) derive some portion of their income or self consumption from Paddy
 - Poverty and hunger (1.46 lakh BPL family) are very high so farmers have a strong preference to grow paddy as its principle kharif crop
 - Most families grow at least what they require for self consumption, with X% of the total production being consumed on the farm
- Farmer with low land also grow paddy in summer if irrigation available**
 - For those with water, paddy can be an attractive summer crop as well – as it is non perishable and easy to sell vs. vegetable which required more labor, is perishable, and requires market visits 2-3 times in week during harvesting
 - Ideal sowing time is usually delayed for other crops like wheat and gram

Kirtti – What is the 100% acreage in the chart? How much is consumed on farm... need to complete XXs

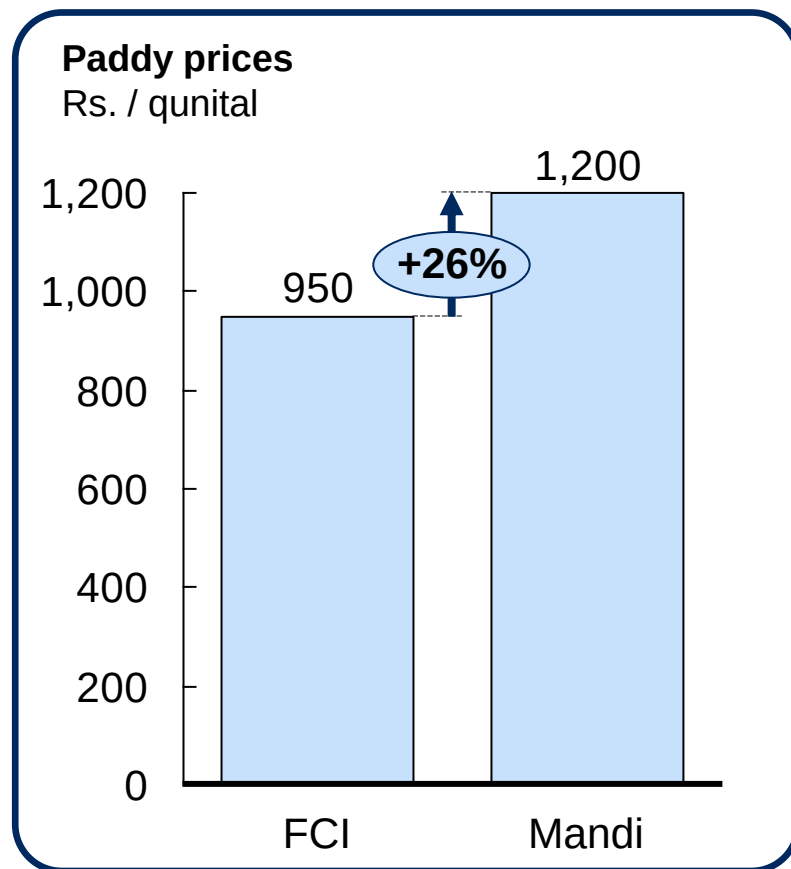
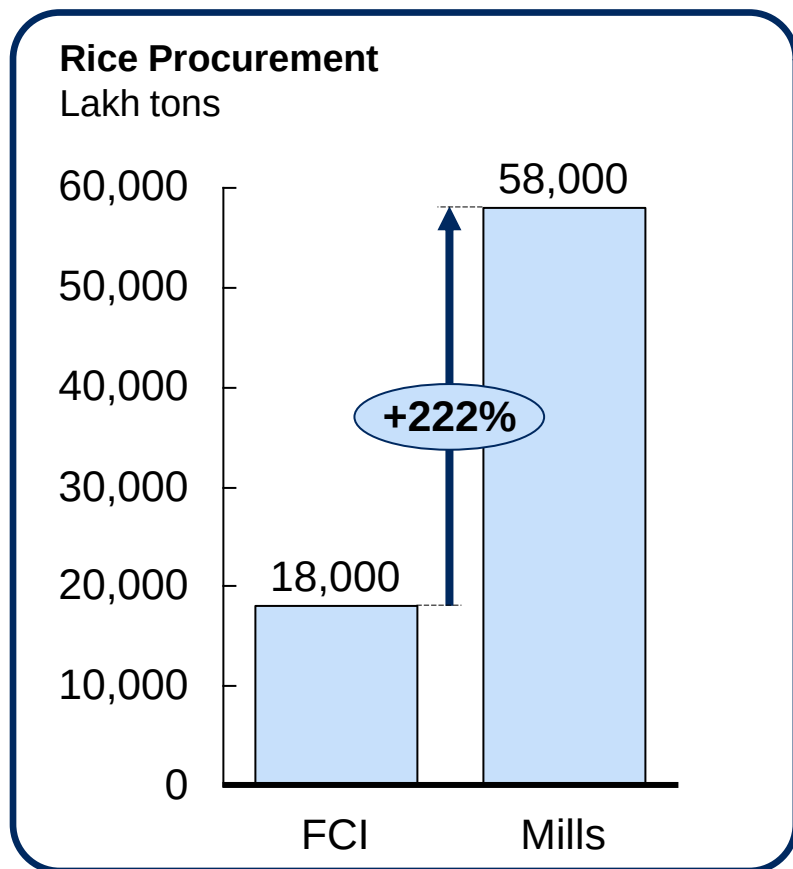
Of the existing paddy farmers, 75% are food insecure and remaining 25% have surplus production

Months of food security in Balaghat
households



- Lack of food security of paddy farmers had limited willingness of farmers to invest and kept productivity far below its potential
- Limited surplus production has also limited the market development, with substantial portions of local rice being imported from places like Bihar
- Many women currently migrate to places like Nagpur and Raipur after paddy harvesting, however this could be reduced if paddy yields achieved their potential

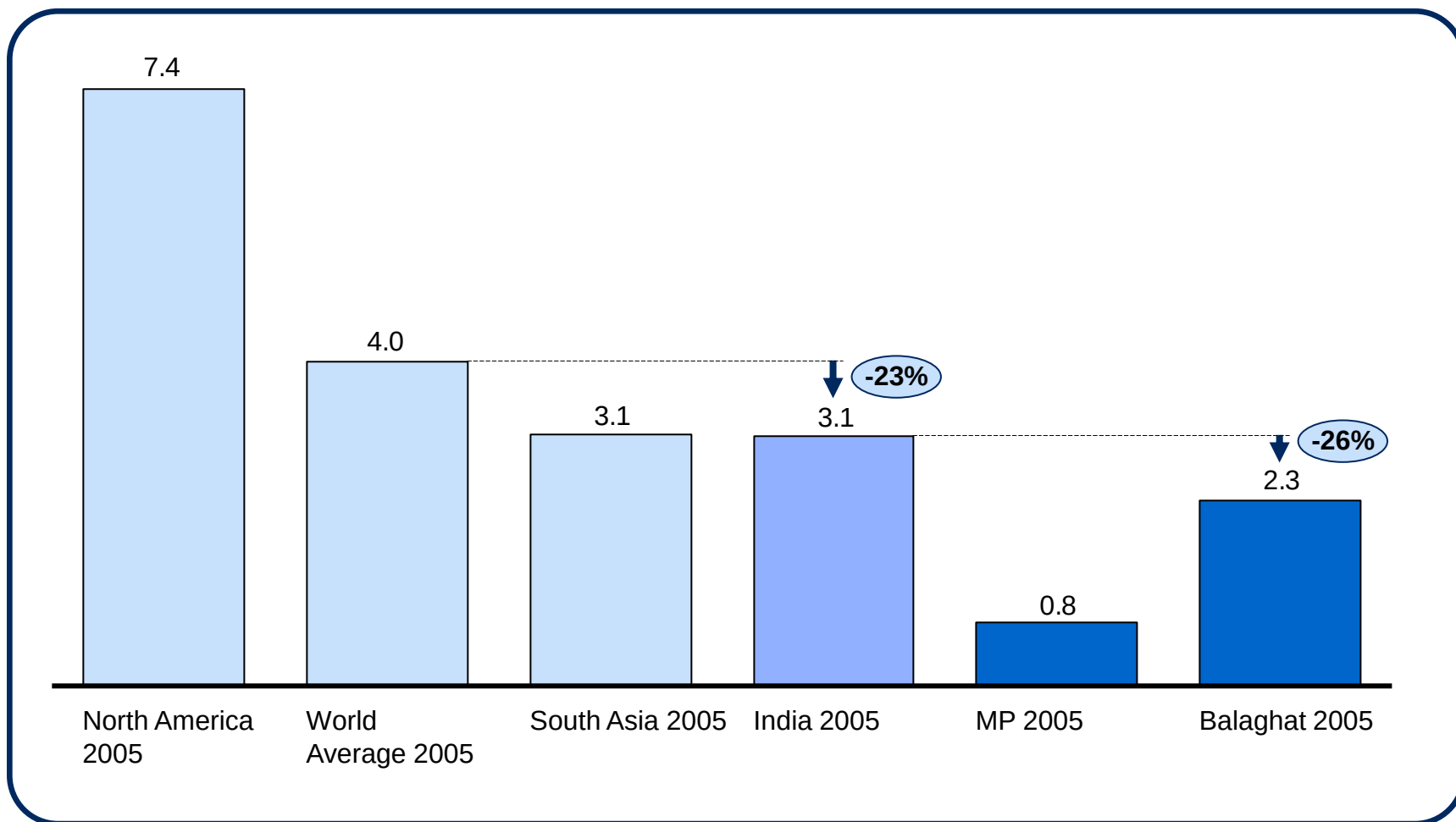
Balaghat also has a competitive paddy market so farmers are able to get higher prices than from state procurement agencies for fine rice



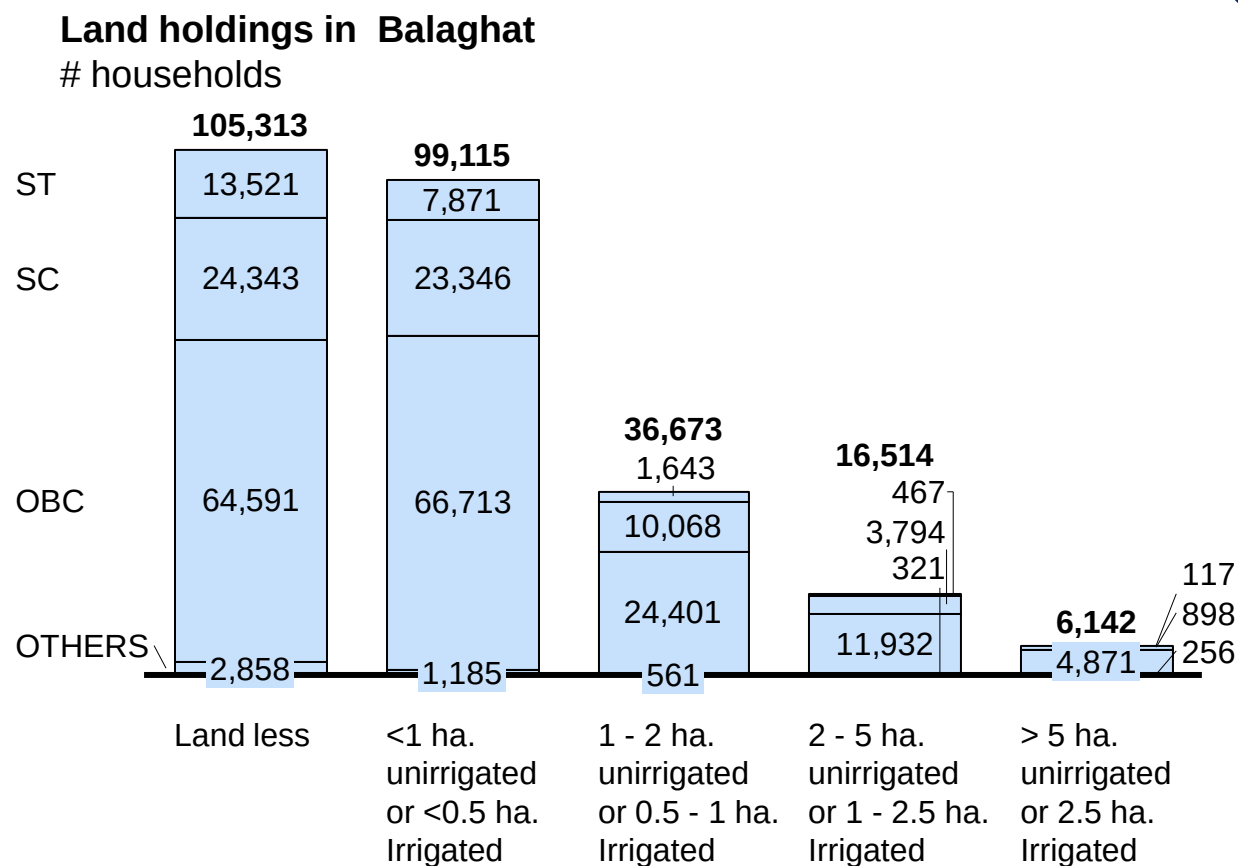
- FCI and cooperative procured mainly Coarse rice and they pay Rs 50 more for fine rice, whereas private market is more focused on fine rice.

In spite of its potential, Balaghat is achieving only 25% of the Indian average yields, and substantially below global average

Paddy Yields - Ton/Ha

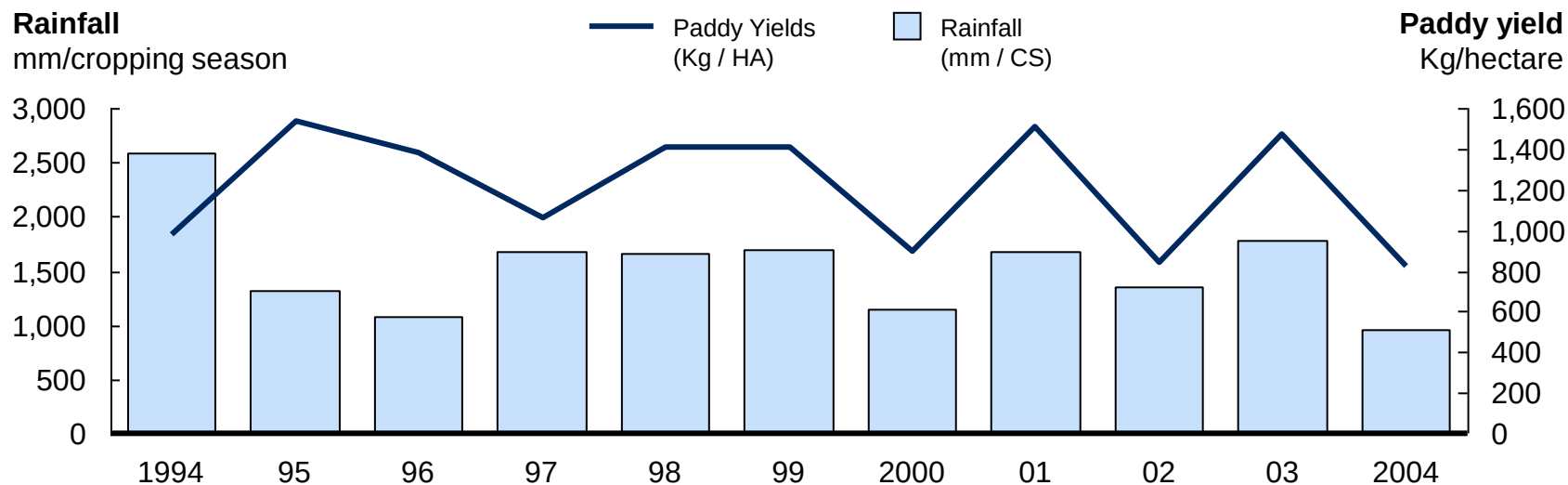


Small landholdings and high poverty result in low willingness to invest and ability to access updated knowledge of practices



- Over 75% of families in Balaghat are marginal or landless and cultivate for self consumption
- As a result, risk taking ability and investment are low contributing to poor yields

Yield also suffers due to erratic rainfall



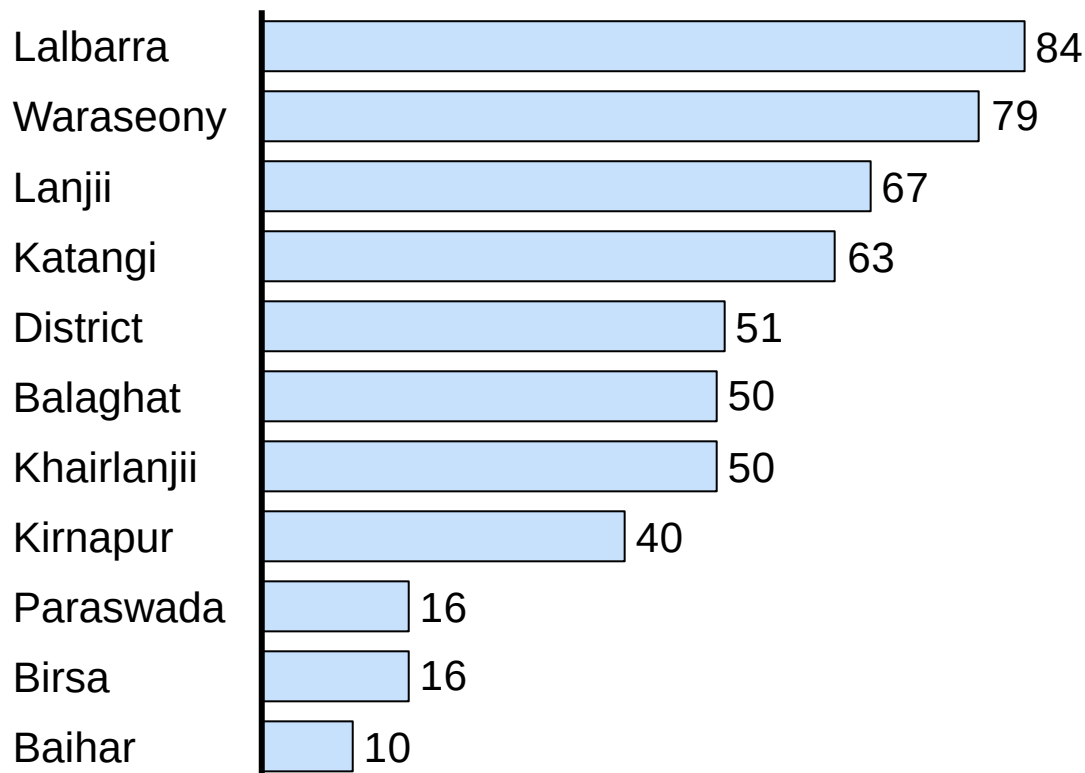
- The normal Annual Rainfall in Balaghat is XXX mm, of which ~X% or ~1,500 mm typically falls within the kharif cropping season
- There is, however, significant volatility (both excess rains such as in 1994-5 or too little such as 2004-5) which reduces both farmers willingness to invest and overall productivity
- Every third year there is large scale crop loss due to prolong dry spells (15-30 days receiving only about 50 to 100 mm of rain) during July to October
- To compensate, farmers grow 120-135 days variety and take risk of water scarcity at time of ripening; Growing of short duration varieties up to 100 days may help reduce vulnerability

Kirtti – Add the average rainfall number in the text comments

The impact of erratic rains is lessened by comparatively high irrigation levels in the district, however some blocks such as Paraswada have been severely neglected

Area under irrigated paddy by block, Balaghat

Percentage of paddy land with irrigation



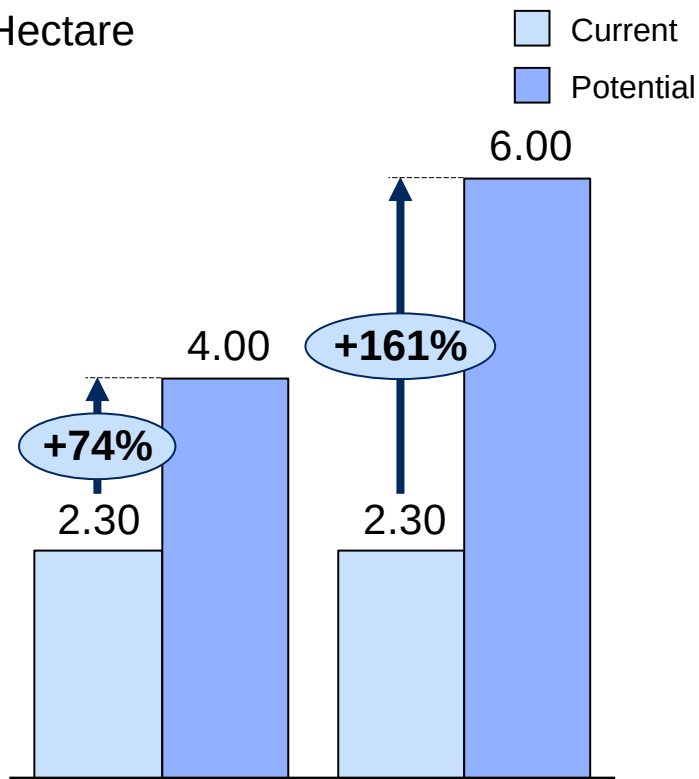
- Overall paddy lands under irrigated is very high at 51%
- However tribal blocks have been severely neglected and have less than 20% irrigation
 - Paraswada (16%)
 - Birsa (16%)
 - Baihar (10%)
- 60% of total irrigation is by canal which is only usable in kharif – alternate forms should be explored

Small and marginal Farmers ability to invest is also limited by poor access to credit

If farmers were able to address these productivity constraints, they have the potential to increase paddy production by XXXX MT, shifting from a deficit to a surplus, meeting food need and increasing incomes by Y%

Paddy Yields in Balaghat

MT / Hectare



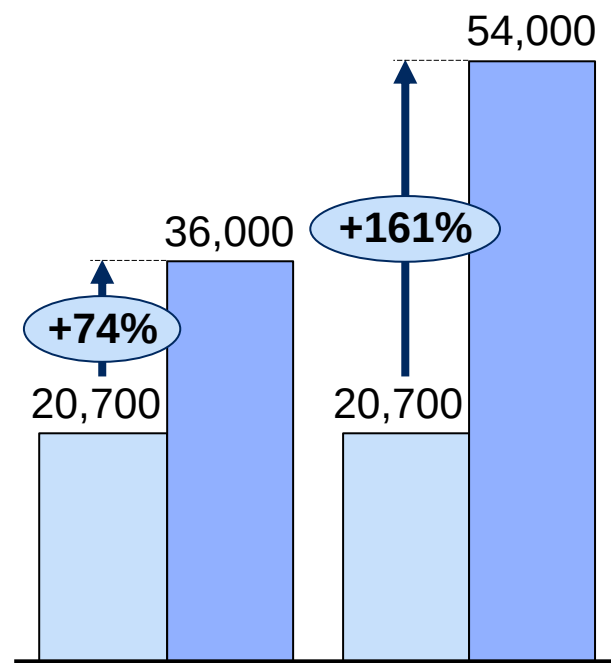
Traditional
Paddy vs
improved

Traditional vs
SRI

Complete the numbers in the header

Paddy Profitability in Balaghat

Rs.000 / Hectare



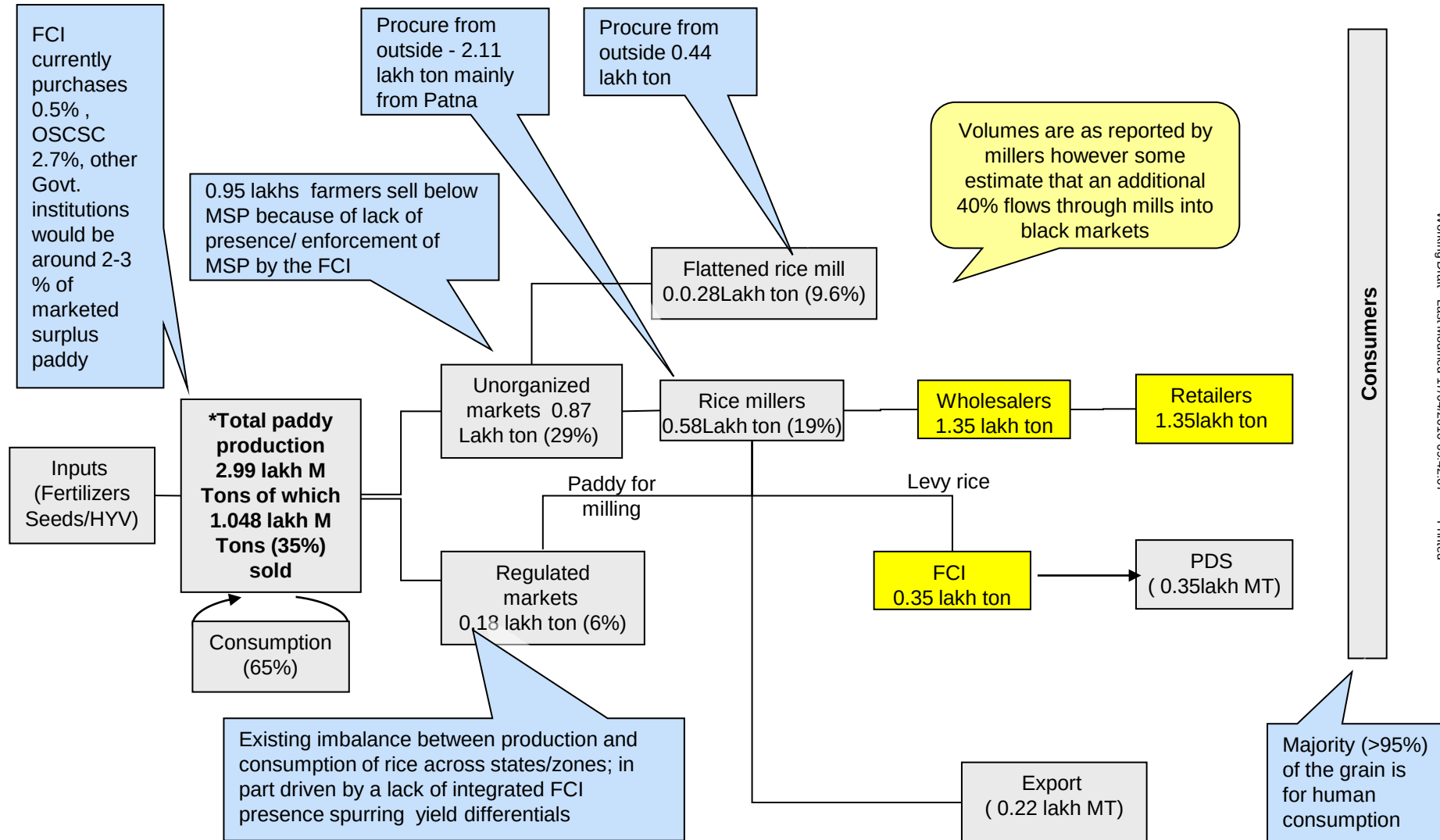
Traditional

Traditional vs
SRI Paddy

Today's discussion

- Why is Paddy a critical crop for both India and Keonjhar?
- What are the current challenges and potential end state?
- **What are the potential interventions?**
- What should we do next?

The current value chain is dominated by private rice mills and wholesalers



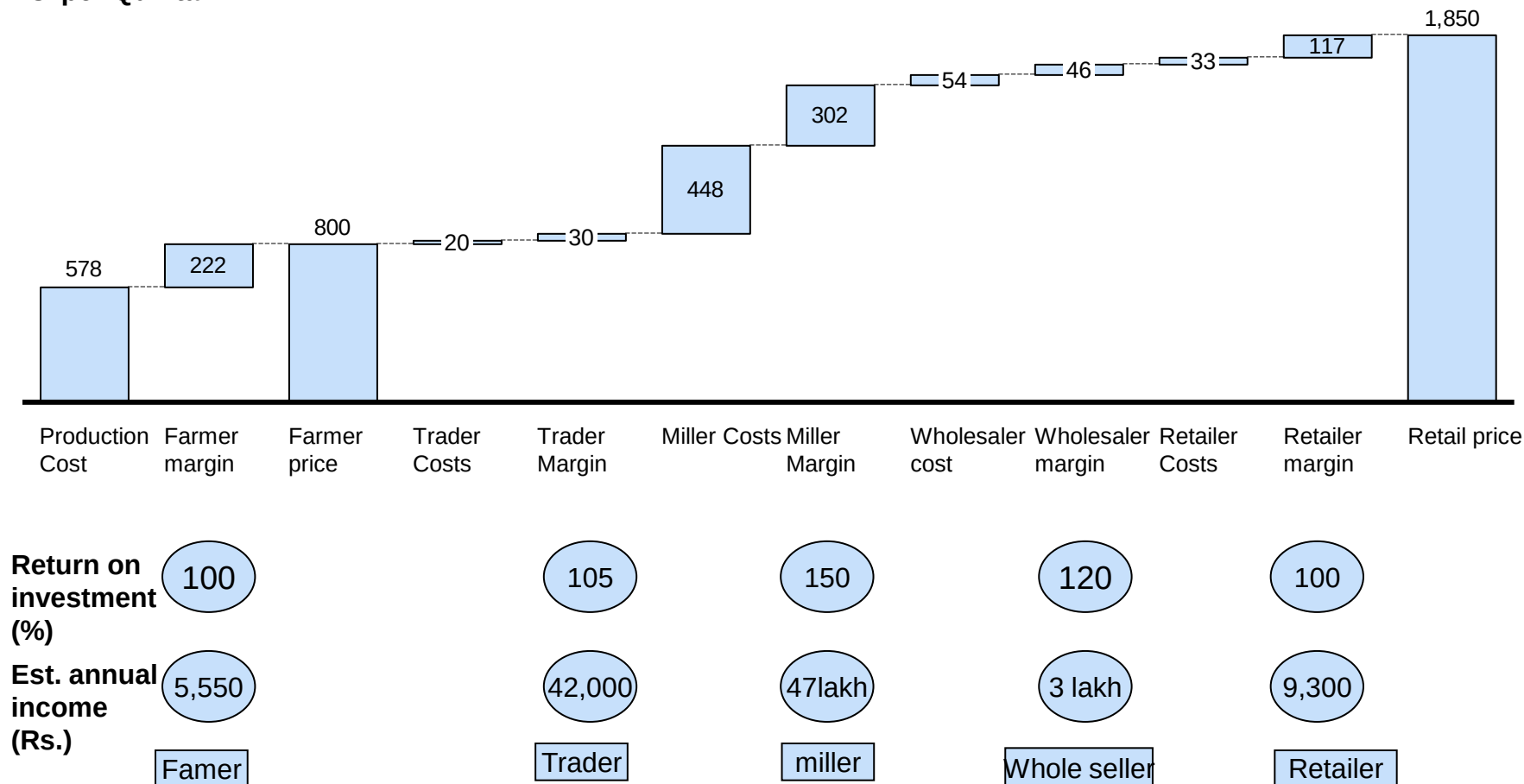
* Assumptions: Yield data is from Agri dept. 05-06; 14.98 lakhs population; per capita rice consumption per year is 1.83Q; Paddy requirement is 3Q; Total Food consumption 4.49 lakhs MT; Present food deficit (1.5 lakh MT); Though many families have less land holding and food deficit; For the whole district 1.04 lakhs MT paddy is sold; Paddy flow from outside is more than 2.55 lakhs ton.

Substantial imports are required from Bihar to meet regional demand



Current Paddy value chain economics in Balaghat

Rs. per Quintal



*Farmers harvest one crop with 25 Q productivity with Rs 222/ Q margin

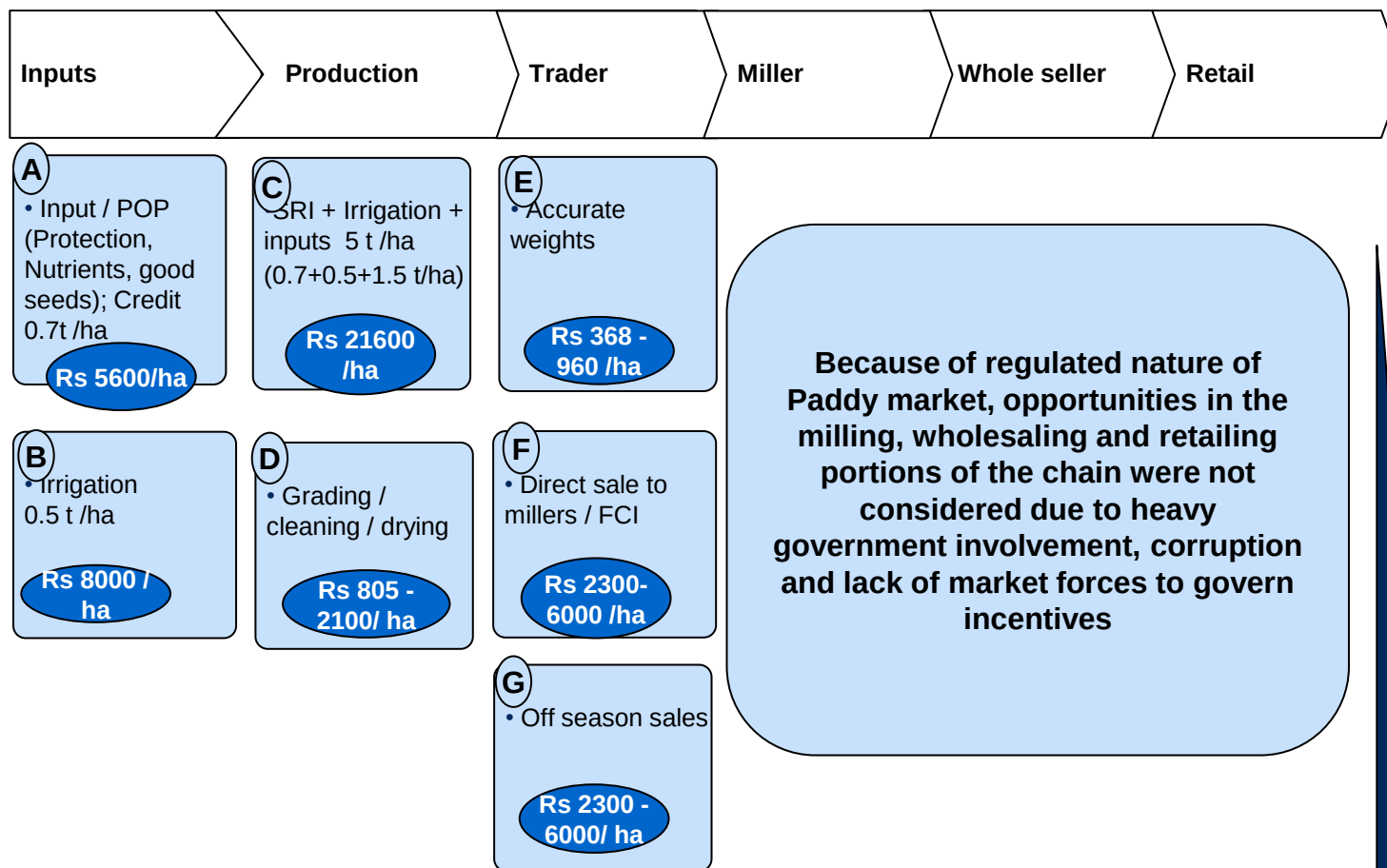
*Miller deals with 15600 q rice and get Rs 302 / Q margine in rice

*Retailer deals with 80Q rice with Rs 117/ Q margin

*Trader deals 1400 Q paddy with margin of Rs 30/ q

* WSD deals with 6500 Q rice with margin of Rs46/ q

We have identified 7 opportunities to create or shift value in the paddy chain



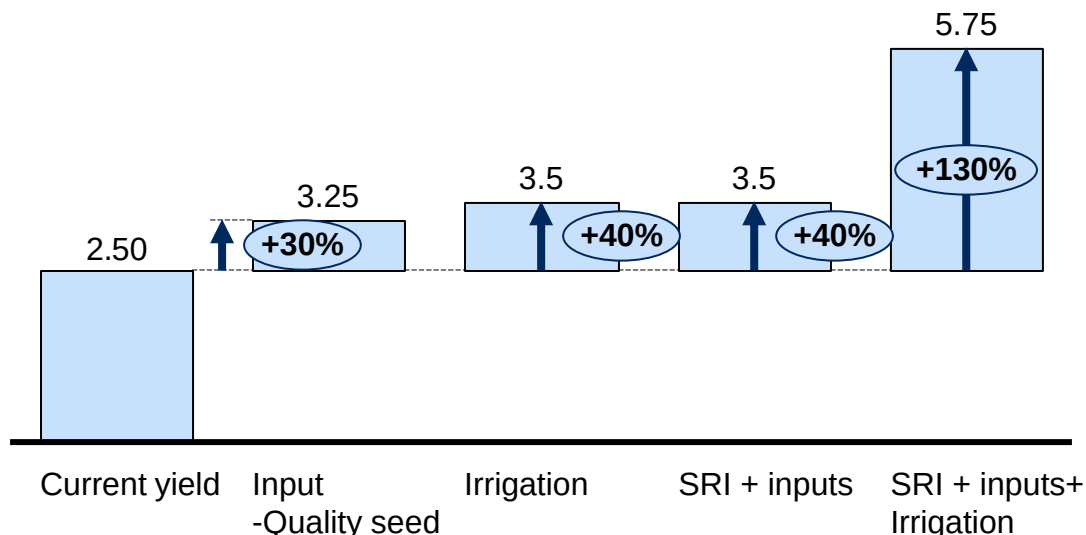
• Existing income is Rs 18,400/ha. Thus this represents a cumulative potential to increase income by up to Rs 46233/ha means Rs 27,833 additional income / ha (251%)

Existing income and potential here doesn't match number on page 18 = please harmonize – add optimizing variety or is it part of POP??

- A** There are opportunities to improve yields by up to 130%, which would
B bring Balaghat up to 200% of the national average

C

Tons/ha



Per hectare

	Current yield	Input -Quality seed - Nutri & Pest	Irrigation	SRI + inputs	SRI + inputs+ Irrigation
Additional output (tons)	2.3	0.75	1	1	3.25
Additional investment	14,462	1,500	3,000**	2,500	5,500
Profit (Rs.)***	3,938				

- SRI with improved inputs is appear as a better opportunity – for this need to aware about SRI techniques and need to link with credit
- For improved input opportunity to link with credit and need to form collective body to input procurement
- Opportunity to create irrigation infrastructures – like rain water harvesting structures and well with pump

Make sure these are consistent v
page 18

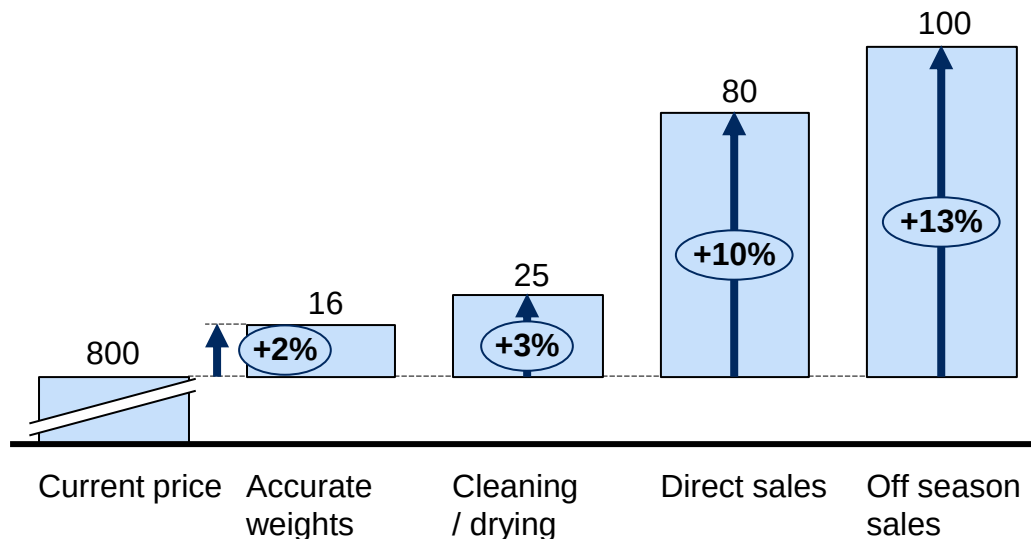
In comments – can we discuss why
these things arent happening today?

* After the additional investment for

** Assuming that irrigation infra
the current local price

- D** While 4 price related initiatives have the potential to increase prices by
E up to Rs. 100 / quintal
F **G**

Rs. / Quintal



Per hectare

	Current price	Accurate weights	Cleaning / drying	Direct sales	Off season sales
Additional price (Rs/Q)	800	16	35	100	100
Additional investment	578	0	16	20	32
Profit (Rs.)***	222	16	19	80	

- Awareness creation about weighing
- awareness about profit of cleaning and grading
- Awareness about procurement centers and pressure to timely payment
- Awareness about price fluctuation and storage infrastructure creation

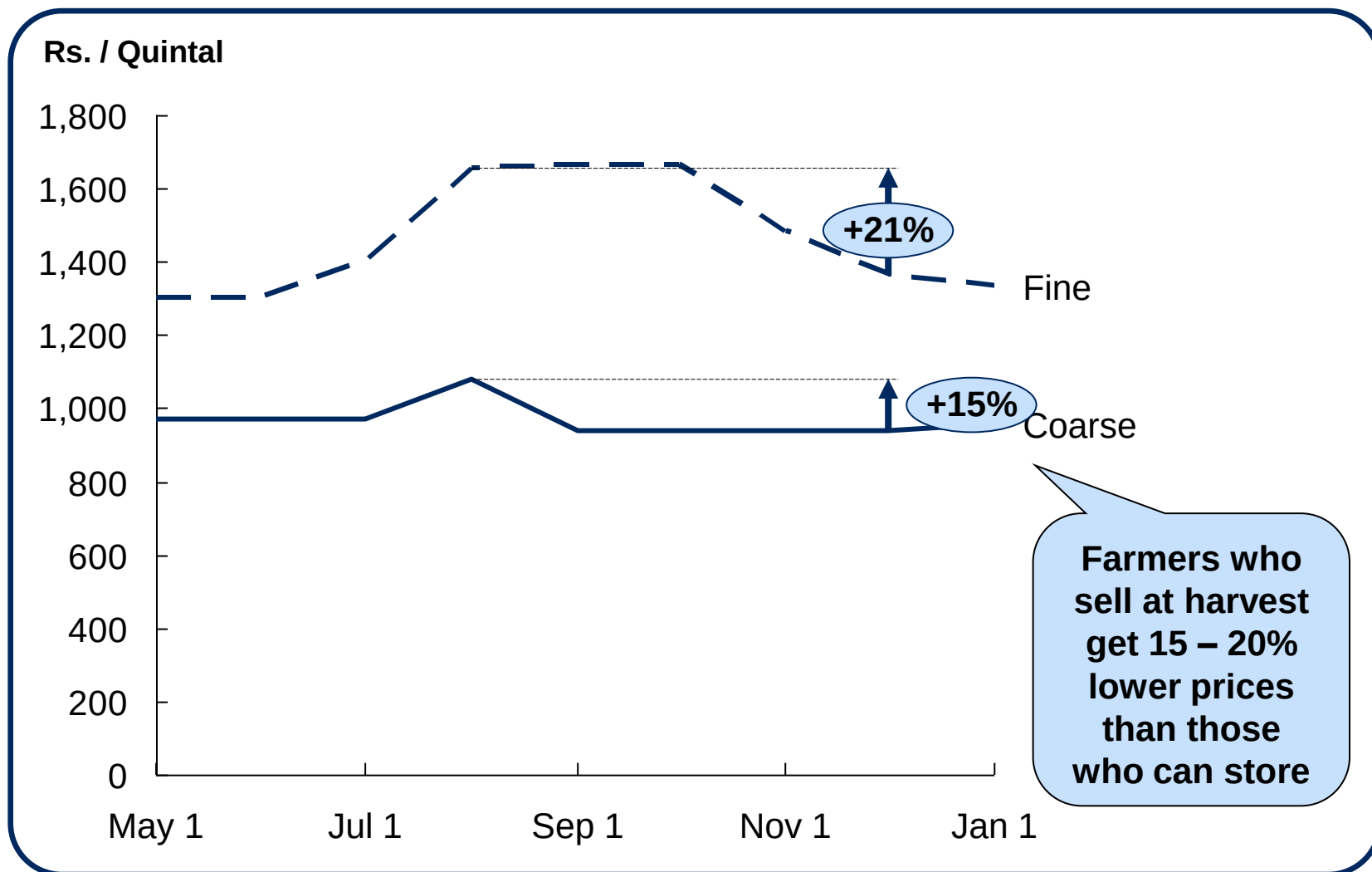
**In comments – can we discuss why these things arent happening today?
Is awareness the only issue??**

* Av. loss occurred in weight up to 2 % if clean properly but get Rs 35 more in sell

** Av Rs 20 transporting & loading un loading charge if sell at procurement centre

*** Av moisture and storage loss 4% if stored up to off season

If farmers sell during the off season to millers, we estimate they could improve their profitability by Rs 100-200/Q



X Optimizing varieties could improve farmer profitability by up to Rs. 7000 / ha

	Profit 000 Rs / HA	Yield Q/ HA	Price / KG	Smallholder / land type suitability
MTU1010	27	30	900	• Coarse grain, upland variety, maturity 110days
Om 3	33	25	1300	• Fine grain, upland variety maturity 100 days
IR64 & 36	32	35	900	• Coarse grain, med upland variety maturity 120 days
Sarthii	39	30	1300	• Fine grain, med upland variety, maturity 125 days
Sarthii	54	60	900	• Coarse grain, low land variety, maturity 135-140 days
Jai sri Ram	70	50	1400	• Fine grain, low land variety, maturity 135-140 days

All fine variety is suitable for market due to high price
For poor farmer coarse give more yield and more food security

8 interventions can be pursued to capture over Rs. 24,000 value for farmers

Opportunity	Yield Impact (additional)	Income Impact	Possible interventions
	MT/Hectare	Rs./Hectare	
A Increase yields through better inputs	0.75	4500	- Collective input procurement - Credit linkage
B Increase Irrigation	1	6500	- Irrigation infrastructure - Rain water harvesting
C SRI + Better Inputs	1.75	14000	Collective input procurement , credit linkage, SRI training & demonstration
D SRI + better input + irrigation	3.5	21500	Collective input procurement , credit linkage, SRI training & demonstration, Irrigation infrastructure & rain water harvesting
E Direct sales to market	N/A	800	Awareness generation and SHG operated procurement centre
F Cleaning, drying, and grading	N/A	250	Awareness generation; introduce winnower
G Sell in off season	N/A	1000	Awareness generation; Storage facility development at HH level
H Transparent weighing system	N/A	160	Introduction of electronic machine at Mandi & procurement centre

The relevance and impact of these interventions varies by segment

Opportunity	1 Market oriented Farmers	2 Subsistence low land	3/4 Subsistence med/up land	5 Marginal med/up land
Representative family	Liladhar 10acre	Nainsingh 4 acre	(Raje singh 4 acre)	(Jhamsingh 1.5 acre)
Objective	▪ Max. productivity and link to market	▪ Max. productivity	Max. productivity and explore other crops	▪ Max. productivity and transition to other crop
Impact	▪ Rs. 1,03,690 / year	▪ Rs. 45830 / year	▪ Rs. 41200 / year	▪ Rs. 14400 / year
A Increase yields through better inputs	✓ additional 27Q Rs 21600	✓ additional 15Q Rs 12000	✓ additional 19Q Rs 15200	✓ additional 6Q Rs 4800
B Increase Irrigation	✓ addition 22Q Rs 17600	✓ additional 8Q Rs 6400	✓ additional 8Q Rs 6400	✓ additional 3 Q Rs 2400
C SRI + Better Inputs	✓ additional 32 Rs 25600	✓ additional 31 Q Rs 24800	✓ additional 39Q . Rs 31200	✓ additional 13Q Rs 10400
D SRI + better input + irrigation	✓ additional 115Q . Rs 92000	✓ additional 55 Q Rs 44000	✓ additional 51Q Rs 40800	✓ additional 18 Q Rs 14400
E Direct sales to market	✓ Rs 5600	✓ Rs 800	✓ Rs 400	X
F Cleaning, drying, and grading	✓ Rs1330	X	X	X
G Sell in off season	✓ Rs			
H Transparent weighing system	✓ Rs			

These segment numbers need to be updated based on new numbering that you did – then just check number to make sure they are consistent with earlier calculations

Today's discussion

- Why is Paddy a critical crop for both India and Balaghat?
- What are the current challenges and potential end state?
- What are the potential interventions?
- **What should we do next?**

Additional topics for future exploration and next steps

**Are these really all the questions? I don't understand the second one...
What about an advocact agenda around FCI?**

QUESTIONS

- Explore rice sub markets; We did not explore flattened rice
- Does paddy seed production is a viable opportunity in Balaghat where market penetration is more?

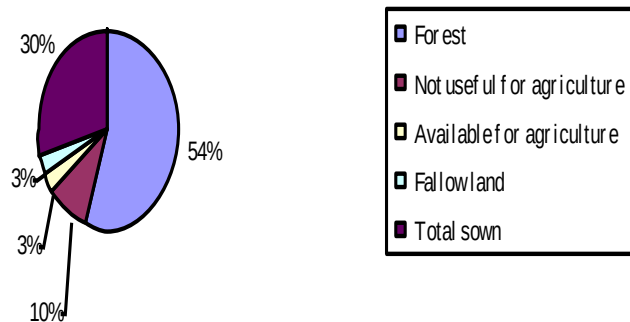
ACTION ITEMS

- Engage community and stakeholders on data to develop roadmap to develop the sector
- Design interventions

Additional information.....

There is also some scope to increase acreage however it is limited

Land resource Balaghat district



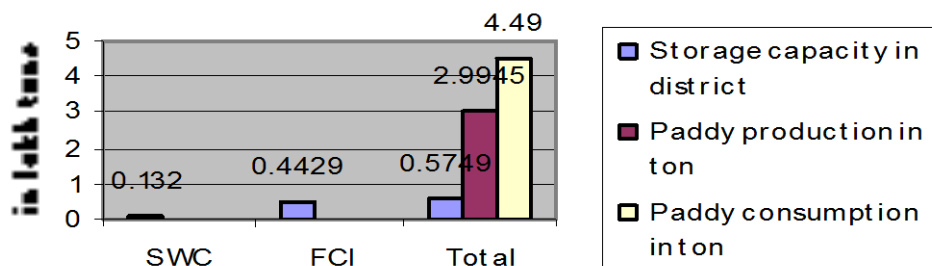
- High forest density with tribal population
- Agriculture occupied only 30% of total land
- No opportunities to increase agriculture land
- Single cropped area

Forest	505004	55
Not useful for agriculture	88606	10
Available for agriculture	28411	3
Fallow land	29663	3
Total sown	272816	30
Total area	924500	100

Source district statistical book 2008

CONSUMPTION

Storage capacity Vs consumption & production of paddy in district



• Storage capacity as compare to total consumption of is less so that for drought year district is not able to store sufficient grain.

• Production is six time high than the storage capacity

Storage capacity of FCI & State warehouse in district in tones			
Location	SWC	FCI	Total
Balaghat	0	20310	20310
Katangi	1500	13980	15480
Lalbarra	1000		1000
Waraseoni	7700	10000	17700
Baihar	2000	0	2000
Langi	1000	0	1000
Storage capacity in district	13200	44290	57490
Paddy production in ton			299450
Paddy consumption in ton			449000
Source : Fertiliser Association of India.			

Revenue per ha with different intervention

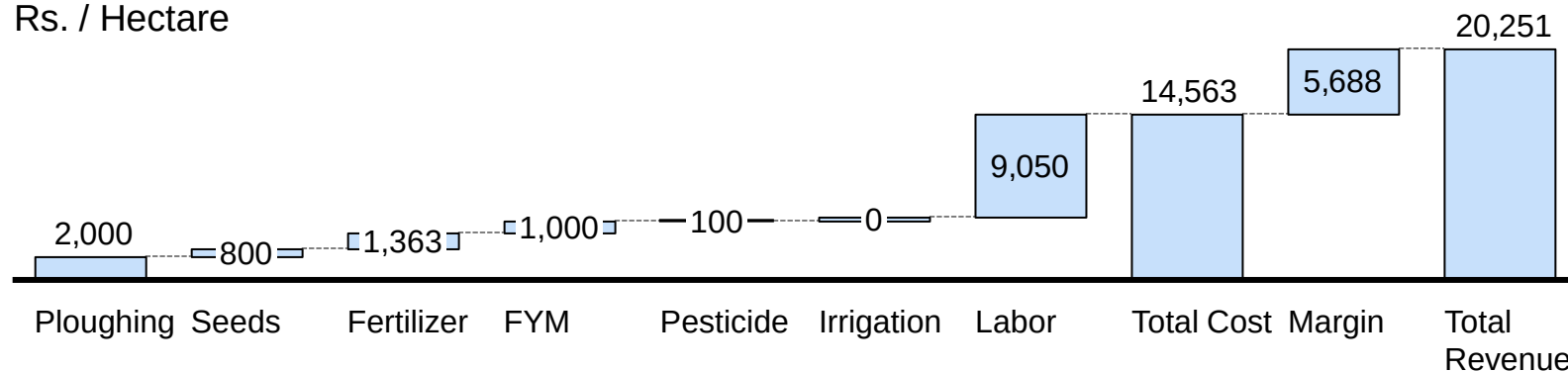
Cost of Rs / ha

List of inputs	Traditional	SRI + better input+ irrigation	with better inputs	only irrigation input	SRI +input
Ploughing	2000	2000	2000	2000	2000
Seeds	800	150	500	800	150
Fertilizers	1362.5	2000	2000	1362.5	2000
Pesticides	100	500	500	100	500
Irrigation	0	1500	0	1500	0
Labour	9050	9050	9050	9050	9050
FYM	1000	1800	1000	1000	1800
weeder	0	500	0	0	500
Total cost	14312.5	17500	15050	15812.5	16000
Total revenue	20000	48000	28000	24000	40000
Margin	5687.5	30500	12950	8187.5	24000

IF FARMER DO PADDY CULTIVATION WITH SRI, IRRIGATION AND BETTER INPUTS

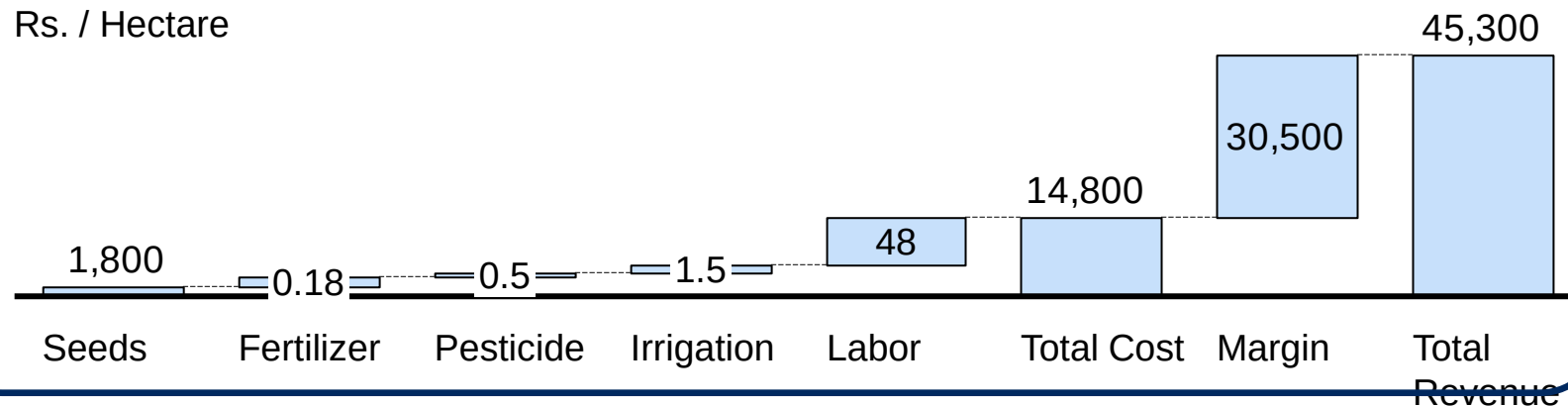
Traditional Paddy Cultivation

Rs. / Hectare



SRI Paddy Cultivation

Rs. / Hectare



Of the existing paddy farmers, 75% are food insecure and remaining 25% have surplus production

		Impact on different categories of farmers				
	Impact on income	< 6 month food security Paddy Land 0.5ha	6-10month food security Paddy Land 1ha	10-12 month food security Paddy land 1.5ha	>12 month food sufficiency Paddy land 4ha	>12 month food sufficiency Paddy land 9 ha
No of farmers	-	99115		36673	16514	6142
Increase yields through better inputs /ha	4500	2250	4500	67.5	18000	40500
Improved by Irrigation /ha	6500	3250	6500	9750	26000	58500
SRI + better Input/ ha	14000	7000	14000	21000	56000	126000
SRI + better input + irrigation	21500	10750	21500	32250	86000	193500
Sell in FCI/ cooperative per ton	800	Not sell	240 (sell0.3 ton)	800 Sell 1ton	5600 Sell 7 ton	16000 Sell 20ton
Cleaning drying grading / ton	250	Not sell	75 (sell0.3 ton)	250 Sell 1ton	1750 Sell 7 ton	5000 Sell 20ton
Sell in off season / ton	1000	Not sell	300 (sell0.3 ton)	1000 Sell 1ton	7000 Sell 7 ton	20000 Sell 20ton
Transparent weighing system / ton	160	Not sell	48 (sell0.3 ton)	160 Sell 1ton	1120 Sell 7 ton	3200 Sell 20ton